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Student ID:

EIGENVECTORS, PCA AND SVD PROBLEM SHEET

1 In this question we consider real symmetric matrices. Throughout, $\mathbf{M}$ is an $n \times n$ real symmetric matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (repeated eigenvalues being listed according to their multiplicity).

Recall if v is an eigenvector then $\mathbf{M}v = \lambda v$ or $(\mathbf{M} - \lambda \mathbf{I})v = \mathbf{0}$. As all vectors in the direction v are mapped to $\mathbf{0}$ the volume of space after applying $(\mathbf{M} - \lambda \mathbf{I})$ is zero. Consequently, $\det(\mathbf{M} - \lambda \mathbf{I}) = 0$. This is the characteristic equation that allows us to find λ . Using the values of λ we have found we can then solve $(\mathbf{M} - \lambda \mathbf{I})v = \mathbf{0}$ for v .

(a) Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and verify that the eigenvectors are orthogonal.

[3 marks]

- (b) Show that the eigenvalues of $\mathbf{M}$ are real, and that two eigenvectors whose eigenvalues differ are necessarily orthogonal. (Hint, for the first part consider $\mathbf{v}^* \mathbf{M} \mathbf{v}$ where $\mathbf{v}^* = \bar{\mathbf{v}}^T$ is the complex-conjugate transpose.) [4 marks]

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4

- (c) Let $\mathbf{V}$ be the matrix whose columns are the normalised eigenvectors of $\mathbf{M}$ and let $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Explain why $\mathbf{V}$ is orthogonal, show that $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, and deduce an expression for $\mathbf{M}^k$. [3 marks]

[illegible]

3

(d) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$, show that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i.$$

[5 marks]

5

(e) Using the cyclic property of the trace, $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$, show that

$$\text{tr } \mathbf{M} = \sum_{i=1}^n \lambda_i.$$

[5 marks]

5

End of question 1

(a) $\frac{1}{3}$	(b) $\frac{1}{4}$	(c) $\frac{1}{3}$	(d) $\frac{1}{5}$	(e) $\frac{1}{5}$	Total	$\frac{1}{20}$
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2 Let $\mathbf{X}$ be a $d \times m$ matrix whose m columns are (mean-subtracted) data vectors of dimension d , scaled so that the covariance matrix is $\mathbf{C} = \mathbf{X}\mathbf{X}^T$. We also define the *dual matrix* $\mathbf{D} = \mathbf{X}^T\mathbf{X}$.

- (a) Show that $\mathbf{C}$ is symmetric and positive semi-definite. Under what condition is it positive definite? [3 marks]

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- (b) The first principal component is the unit vector $\mathbf{w}$ maximising the variance $\mathbf{w}^T\mathbf{C}\mathbf{w}$ of the projected data. Using a Lagrange multiplier to impose $\|\mathbf{w}\| = 1$, show that $\mathbf{w}$ must be an eigenvector of $\mathbf{C}$, and identify the value of the Lagrange multiplier. [5 marks]

5

- (c) Let $v_1, \dots, v_d$ be the normalised eigenvectors of $\mathbf{C}$ ordered by decreasing eigenvalue and let $\mathbf{P} = \sum_{i=1}^k v_i v_i^\top$. Show that $\mathbf{P}^2 = \mathbf{P}$ and that the expected squared reconstruction error $\mathbb{E}[\|\mathbf{P}\mathbf{x} - \mathbf{x}\|^2]$ is $\sum_{i>k} \lambda_i$. [3 marks]

3

- (d) Show that if v is an eigenvector of $\mathbf{D}$ with non-zero eigenvalue λ then $\mathbf{X}v$ is an eigenvector of $\mathbf{C}$ with the same eigenvalue. Why must we exclude $\lambda = 0$? Why is this result useful when there are many more features than examples? [5 marks]

5

- (e) The singular value decomposition writes $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal and $\mathbf{S}$ is diagonal with non-negative entries s_i . Show that

$$\mathbf{X}\mathbf{X}^T = \mathbf{U}\mathbf{S}\mathbf{S}^T\mathbf{U}^T, \quad \mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T$$

and hence say what the columns of $\mathbf{U}$ and $\mathbf{V}$ are, and how the singular values relate to the eigenvalues. [4 marks]

4

End of question 2

(a) $\frac{1}{3}$	(b) $\frac{1}{5}$	(c) $\frac{1}{3}$	(d) $\frac{1}{5}$	(e) $\frac{1}{4}$	Total $\frac{1}{20}$
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3 In this question I want you to use a numerical package to check the results of the previous two questions. You can use MATLAB, OCTAVE, numpy in Python or any other language of your choosing. The answers are only going to be accurate up to the numerical precision of your machine (typically around 10^{-16}), so two quantities should be treated as equal when they agree to around ten decimal places. For python the following may be useful

<code>import numpy as np</code>	import the numpy module
<code>A = np.random.randn(n,m)</code>	generate a matrix of normally distributed random deviates with n rows and m columns
<code>A.transpose()</code>	return the transpose of A (does not change A)
<code>A@B</code>	matrix multiplication AB . Beware that <code>A*B</code> performs elementwise multiplication which is different to normal matrix multiplication
<code>A.trace()</code>	returns the trace of a matrix
<code>np.linalg.det(A)</code>	return the determinant of a matrix
<code>np.linalg.eigh(A)</code>	eigenvalues and eigenvectors of a <i>symmetric</i> matrix A , returned as a pair (<code>lam</code> , <code>V</code>) where <code>lam</code> is a one-dimensional array of eigenvalues in increasing order and the columns of <code>V</code> are the corresponding normalised eigenvectors
<code>np.linalg.eig(A)</code>	the same for a general square matrix. The eigenvalues come in no particular order and may be complex
<code>np.linalg.svd(X)</code>	singular value decomposition, returned as a triple (<code>U</code> , <code>s</code> , <code>Vt</code>). Note that <code>s</code> is a one-dimensional array of singular values in decreasing order, not a matrix, and that the third element returned is $\mathbf{V}^T$ and not V
<code>np.diag(s)</code>	turn a one-dimensional array into a diagonal matrix (and, given a matrix, extract its diagonal)
<code>np.sort(x)</code>	sort an array into increasing order
<code>np.allclose(a, b)</code>	test whether two arrays agree to within numerical precision. Returns a single true or false value
<code>np.round(A, 10)</code>	rounds the elements in a matrix A to 10 decimal places. Makes expressions like <code>(A@B)@C</code> – <code>A@(B@C)</code> easier to read

Generate a random 5×5 matrix **M** and form the symmetric matrix $\mathbf{A} = \mathbf{M} + \mathbf{M}^T$. Separately generate a random 4×6 matrix **X**.

- (a) Compute the eigenvalues of $\mathbf{A}$. Compare the product of the eigenvalues with `np.linalg.det(A)`, and their sum with `A.trace()`. Which of the results of question 1 does this check? [3 marks]

3

- (b) Verify numerically that $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ and that $\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$, where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_5)$. What happens if you replace $\mathbf{A}$ by the non-symmetric $\mathbf{M}$? [3 marks]

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- (c) Compute the singular value decomposition of $\mathbf{X}$. Check that the singular values are the square roots of the non-zero eigenvalues of both $\mathbf{X} \mathbf{X}^T$ and $\mathbf{X}^T \mathbf{X}$. [4 marks]

4

- (d) Check that the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{XX}^T$ and the columns of $\mathbf{V}$ are eigenvectors of $\mathbf{X}^T\mathbf{X}$. You will probably find that some columns come back with the opposite sign to the ones you computed in the previous part. Why is this not an error? [4 marks]

4

- (e) $\mathbf{X}$ is 4×6 , so one of $\mathbf{XX}^T$ and $\mathbf{X}^T\mathbf{X}$ is necessarily singular. Which, and why? Confirm this from the eigenvalues and from the determinant. [3 marks]

3

- (f) Form the rank- k approximation $\mathbf{X}_k = \sum_{i=1}^k s_i \mathbf{u}_i \mathbf{v}_i^T$ for $k = 1, 2, 3, 4$ and compare the squared error $\sum_{jl} (X_{jl} - (X_k)_{jl})^2$ with $\sum_{i>k} s_i^2$. [3 marks]

3

End of question 3

(a) $\frac{1}{3}$	(b) $\frac{1}{3}$	(c) $\frac{1}{4}$	(d) $\frac{1}{4}$	(e) $\frac{1}{3}$	(f) $\frac{1}{3}$	Total $\frac{1}{20}$
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4 Almost everything we have done so far assumed the matrix was real and symmetric, which guaranteed real eigenvalues and a complete set of orthogonal eigenvectors. This question is about what happens when that assumption is dropped. The point is not that the calculations are harder, but that the conclusions genuinely change. You may use python throughout.

(a) By hand, find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Show that $\mathbf{A}$ has a repeated eigenvalue but only one eigenvector direction, and explain why this means $\mathbf{A}$ cannot be written as $\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$. [4 marks]

4

(b) Confirm this in python using `np.linalg.eig`. You will get back a matrix with two columns: look at them and explain what has happened. Compute the determinant of that matrix and say why $\mathbf{V}^{-1}$ cannot be formed.

[3 marks]

3

(c) Now take

$$\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find its eigenvectors and show that, although they are not orthogonal, $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ still holds. What does this tell you about the difference between *non-symmetric* and *defective*?

[3 marks]

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(d) Consider the rotation matrix

$$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Compute its eigensystem numerically for $\theta = \pi/3$. Verify that the eigenvalues are $e^{\pm i\theta}$ and that the two eigenvectors are complex conjugates of each other. Explain why complex eigenvalues of a *real* matrix must come in conjugate pairs. What replaces the sign ambiguity of question 3(d) in the complex case?

[4 marks]

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(e) Check that $\det(\mathbf{R}) = \prod_i \lambda_i$ and $\text{tr } \mathbf{R} = \sum_i \lambda_i$ still hold, even though the eigenvalues are complex. Why do the answers come out real?

[3 marks]

3

- (f) Finally, compute the singular value decomposition of the defective matrix $\mathbf{A}$ of part (a). It exists, and its singular values are real and positive, even though $\mathbf{A}$ has no eigendecomposition. Explain why, and confirm that $s_1 s_2 = |\det(\mathbf{A})|$.
[3 marks]

3

End of question 4

(a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$ (e) $\frac{1}{3}$ (f) $\frac{1}{3}$ Total $\frac{1}{20}$